

ON THE NUMBER OF SETS OF CYCLE LENGTHS

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A set S of integers is called a *cycle set* on $\{1, 2, \dots, n\}$ if there exists a graph G on n vertices such that the set of lengths of cycles in G is S . Erdős conjectured that the number of cycle sets on $\{1, 2, \dots, n\}$ is $o(2^n)$. In this paper, we verify this conjecture by proving that there exists an absolute constant $c \geq 0.1$ such that the number of cycle sets on $\{1, 2, \dots, n\}$ is $o(2^{n-n^c})$.

1. Introduction

For any graph G , the set of lengths of cycles in G is denoted by $C(G)$. A set S of integers is called a *cycle set* on $\{1, 2, \dots, n\}$ if there exists a graph G on n vertices such that $C(G) = S$. There are numerous results relating the edge density and arithmetic properties of cycle sets in graphs. Possibly the simplest statement is that if G is a graph of minimum degree at least k , then $|C(G)| \geq k - 1$. The residue properties of $C(G)$ are also well-researched: for example, Bondy and Vince [3] proved that if G is a graph of minimum degree at least three, then $C(G)$ contains a pair of consecutive integers or a pair of consecutive even integers. The following result of the author [15] partly generalises this: any graph of average degree at least $8k$ contains cycles of k consecutive even lengths. Very general results pertaining to the density of the set of cycle lengths were established by Gyárfás, Komlós and Szemerédi [11]: they proved that there exists an absolute positive constant ε such that for every graph G of average degree at least d , the sum of reciprocals of elements

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of $C(G)$ is at least $\varepsilon \log d$. Although these results suggest that cycle sets on $\{1, 2, \dots, n\}$ differ significantly from ordinary subsets of $\{1, 2, \dots, n\}$, they do not imply the following conjecture of Erdős [5]:

Conjecture 1.1. The number of cycle sets on $\{1, 2, \dots, n\}$ is $o(2^n)$.

[Conjecture 1.1](#) has been studied for certain classes of graphs by Denley [4]. From the results of the author [13], it can be deduced that the number of cycle sets on $\{1, 2, \dots, n\}$ arising from the class of graphs on n vertices and of average degree at least $9 \log_2 \log_2 n$ is $o(2^n)$. In this paper, we will verify [Conjecture 1.1](#) by proving the following theorem:

Theorem 1.2. *There exists an absolute positive constant c such that the number of cycle sets on $\{1, 2, \dots, n\}$ is $o(2^{n-n^c})$.*

A simple lower bound for the number of cycle sets on $\{1, 2, \dots, n\}$ is given approximately by the partition function

$$p(n) \sim \frac{1}{4\sqrt{3n}} e^{\pi\sqrt{2n/3}},$$

by considering the class of graphs on n vertices comprising vertex-disjoint cycles. A more effective example is as follows: let $A \subset \{3, 5, \dots, 2n-3\}$, $C = v_0 v_1 \dots v_{2n-1} v_0$ and define $G_{2n}^A = C + \{v_0 v_i : i+1 \in A\}$. Then $C(G_{2n}^A) \cap (2\mathbb{Z} + 1) = A$. So the number of distinct sets $C(G_{2n}^A)$ is at least 2^{n-2} . A slightly better example was found by Faudree: let $A \subset \{n+1, \dots, 2n-1\}$ and $G_{2n}^A = C + \{v_0 v_{i-1} : i \in A\}$. Then $C(G_A) \cap \{n, n+1, \dots, 2n\} = A$ and therefore the number of distinct sets $C(G_A)$ is at least 2^{n-1} . Let $C(n)$ be the number of cycle sets on $\{1, 2, \dots, n\}$. It would be interesting to determine the limit (if it exists)

$$\lim_{n \rightarrow \infty} n^{-1} \log_2 C(n).$$

The above examples show that the limit is at least $\frac{1}{2}$.

This paper is organised as follows: in [section 2](#), we count the number of sets in $\{1, 2, \dots, n\}$ containing a certain strong additive structure. In [section 3](#), we prove that all but a negligible number of graphs G_n contain subgraphs each of which guarantees that $C(G_n)$ contains this strong additive structure. The proof of [Theorem 1.2](#) is then completed in [section 4](#).

Notation. In this paper, G_n denotes a generic graph of order n , $V(G_n)$ its vertex set and $E(G_n)$ its edge set. A path of length k with vertex set $\{u_1, u_2, \dots, u_{k+1}\}$ is represented by $u_1 u_2 \dots u_{k+1}$. Similar notation is used for cycles. As we have mentioned above, $C(G)$ denotes the set of lengths of cycles

in a graph G , and we write $C(n)$ for the number of cycle sets on $\{1, 2, \dots, n\}$. The notation $\mathbb{Z}_n$ is used for the integers modulo n and $\mathbb{N}$ denotes the set of natural numbers. If S is a set, then its complement is denoted S^c .

2. Counting Sums of Sets

Let A, B be sets of integers and define the *positive difference set* of A and B by $(A - B)^+ = \{a - b : a \in A, b \in B, a > b\}$. The *Fourier transform* of a complex-valued function f on $\mathbb{Z}_{2n+1}$ is:

$$\hat{f}(x) = \sum_{r=0}^{2n} f(r) \exp(2\pi i r x / (2n + 1)).$$

For convenience, we replace $\exp(2\pi i r x / (2n + 1))$ by ω^{rx} . If $A \subset \mathbb{Z}_{2n+1}$, then $A(r) = 1$ if $r \in A$ and $A(r) = 0$ otherwise. References on set summation and properties of Fourier transforms can be found in Nathanson [14] and Freiman [8]. We will count the number of subsets of $\{1, 2, \dots, n\}$ containing large (in a sense made precise later) positive difference sets. If a subset of $\{1, 2, \dots, n\}$ contains such a set, then its Fourier transform is large in modulus at some point in $\{1, 2, \dots, n\}$. The probability that a random subset of $\{1, 2, \dots, n\}$ has a large transform on some point is then estimated. The only non-elementary tool required is the following concentration result derived from Hoeffding's Inequality (see for example McDiarmid's formulation [13], Proposition 7.11, page 175):

Proposition 2.1. *Let $f : \{0, 1\}^n \rightarrow \mathbb{R}$ be such that f changes by at most c when the i th co-ordinate changes. Let $Y = f(X_1, X_2, \dots, X_n)$ where X_i are chosen independently and at random from $\{0, 1\}$. Then $\Pr[Y \geq E[Y] + t] \leq \exp(-t^2 / 2c^2 n)$.*

Lemma 2.2. *Suppose $A, S \subset \{1, 2, \dots, n\}$ and $S \supset (A - A)^+$. Let $T = S^c \cup (-S^c) \pmod{2n+1}$. Then $\max_{r \neq 0} |\hat{T}(r)| \geq |S^c| |A| / n$.*

Proof. As $t = a - a'$ has no solution with $t \in T^c$ and $a, a' \in A$, we have:

$$\sum_{r=0}^{2n} \sum_{t=0}^{2n} \sum_{a=0}^{2n} \sum_{a'=0}^{2n} A(a) A(a') T(t) \omega^{r(t-a+a')} = 0.$$

Therefore $\sum_{r=0}^{2n} \hat{T}(r) |\hat{A}(r)|^2 = 0$. For $r = 0$, the summand is $2|S^c| |A|^2$. So if $m = \max_{r \neq 0} |\hat{T}(r)|$ then

$$2|S^c| |A|^2 = \left| \sum_{r=1}^{2n} \hat{T}(r) |\hat{A}(r)|^2 \right|$$

$$\leq m \sum_{r=1}^{2n} |\hat{A}(r)|^2 = m((2n+1)|A| - |A|^2).$$

Since $|A| \geq 1$, we find $m \geq |S^c||A|/n$ as required. $\blacksquare$

Using [Proposition 2.1](#) and [Lemma 2.2](#), it is relatively straightforward to count the number of sets containing large positive difference sets:

Lemma 2.3. *The number of subsets of $\{1, 2, \dots, n\}$ containing a positive difference set $(A - A)^+$ with $A \subset \{1, \dots, n\}$ and $|A| > a$ is at most $(2n+1)2^{n-a^2/18n}$.*

Proof. Let S be a random set whose elements are chosen uniformly with probability $\frac{1}{2}$ from $\{1, 2, \dots, n\}$. Let $X_i = S(i)$ and $f(X_1, X_2, \dots, X_n) = \sum X_i = |S|$. By [Proposition 2.1](#), $\Pr[|S| \geq 2n/3] \leq \exp(-n/18)$. The transform of T in [Lemma 2.2](#) is

$$\hat{T}(r) = \sum_{i=1}^n (1 - X_i)(\omega^r + \omega^{-r}).$$

Thus $\hat{T}(r)$ is real, $\hat{T}(r) = \hat{T}(2n+1-r)$ and $E[\hat{T}(r)] = 0$. By [Lemma 2.2](#) and [Proposition 2.1](#) applied to $f = \hat{T}(r)$, $\Pr[\hat{T}(r) \geq t] \leq \exp(-t^2/2n)$. By applying [Proposition 2.1](#) to $1 - \hat{T}(r)$, we obtain $\Pr[|\hat{T}(r)| \geq t] \leq 2\exp(-t^2/2n)$. Now define the events

$$E_1 = \{ |S^c| \leq n/3 \},$$

$$E_2 = \{ \exists r \neq 0 : |\hat{T}(r)| \geq n/3 \}.$$

Then $\Pr[E_1] \leq \exp(-n/18)$ and $\Pr[E_2] \leq 2n\exp(-n/18)$. Consequently, the number of sets S in $\{1, 2, \dots, n\}$ containing $(A - A)^+$ for some $A \subset \{1, 2, \dots, n\}$ with $|A| \geq a$ is at most $(2n+1)2^{n-a^2/18n}$. $\blacksquare$

This lemma is effective when $a/\sqrt{n} \rightarrow \infty$. With this in mind, we call a positive difference set $(A - A)^+$ in $\{1, 2, \dots, n\}$ η -large if $|A| \geq \sqrt{18n}^{(1+\eta)/2}$ and $\eta > 0$. By [Lemma 2.3](#), the number of sets $S \subset \{1, 2, \dots, n\}$ containing an η -large positive difference set is $O(2^{n-n^\eta})$.

A more difficult problem, of independent interest, is to determine the asymptotic number of subsets of $\{1, 2, \dots, n\}$ or of $\mathbb{Z}_n$ of the form $A + A$ for some $A \subset \{1, 2, \dots, n\}$. Green and Ruzsa [\[10\]](#) solved this problem by showing that the number of subsets of $\{1, 2, \dots, n\}$ of the form $A + A$ is $\Theta(2^{n/2})$. They also showed that the number of subsets of $\mathbb{Z}_p$ of the form $A + A$ is $\Theta(p^{2p/3})$ when p is prime. This is related problem to the well-known Erdős–Cameron conjecture [\[8\]](#), that the number of sum-free subsets of $\{1, 2, \dots, n\}$

is $O(2^{n/2})$. This problem was studied by many authors, and was finally verified by Green [9].

Let A be a multiset with elements $a_1, a_2, \dots, a_n$. We define the *set of subset sums* of A to be the set

$$A^* = \{\varepsilon_1 a_1 + \dots + \varepsilon_n a_n : \varepsilon_i \in \{0, 1\}, 1 \leq i \leq n\}.$$

A beautiful theorem of Erdős and Sárközy [7] states that if A and A^* are subsets of $\{1, 2, \dots, n\}$ and $|A| \geq 18(\log_2 n)^2$, then A^* contains an arithmetic progression of length $|A|/18(\log_2 n)^2$. We outline their proof, which also works for multisets:

Proposition 2.4. *Suppose A is a multiset with elements $a_1, a_2, \dots, a_t \in \{1, 2, \dots, n\}$ and $t \geq 18(\log_2 n)^2$. Then A^* contains at least $t/18(\log_2 n)^2$ consecutive multiples of some positive integer.*

Proof. Set $q = \lfloor \log_2 n \rfloor$ and $p = \lfloor t/18(\log_2 n)^2 \rfloor$. By the pigeonhole principle, there exists a number m such that at least

$$\frac{1}{qn} \binom{t}{q} > q! p^{q+1}$$

sets of q elements of A sum to m . If $\mathcal{A}$ is the set system comprising those sets $I \subset \{1, 2, \dots, t\}$ of size q such that $\sum_{i \in I} a_i = m$, then, by a result of Erdős and Rado [6], there exists a set system $\mathcal{B} \subset \mathcal{A}$ such that $|\mathcal{B}| > p$, and all intersections of distinct sets in $\mathcal{B}$ are equal. Let B be the common intersection of the elements of $\mathcal{B}$. Then more than p consecutive multiples of $m - \sum_{a_i \in B} a_i$ appear in A^* . ■

We will use the following terminology: a set of subset sums $A^* \subset \{1, 2, \dots, n\}$ is called η -large if $|A| \geq 20n^\eta(\log_2 n)^2$ and $\eta > 0$. The following lemma shows that few sets contain a translate of an η -large set of subset sums:

Lemma 2.5. *The number of sets $S \subset \{1, 2, \dots, n\}$ containing a translate of an η -large set of subset sums is $O(2^{n-n^\eta})$.*

Proof. If A^* is an η -large set of subset sums, then by Proposition 2.4, A^* contains at least $\frac{10}{9}n^\eta$ consecutive multiples of some positive integer. In any translate of A^* , this set corresponds to an arithmetic progression. There are at most n^2 arithmetic progressions in $\{1, 2, \dots, n\}$, as there are at most n choices for the first term and n choices for the common difference. Let S be a random set whose elements are selected uniformly from $\{1, 2, \dots, n\}$

with probability $\frac{1}{2}$. Then the probability that S contains a given arithmetic progression of length k is 2^{-k} . So the number of sets containing such an arithmetic progression is at most $n^2 2^{-k} 2^n$. With $k = \frac{10}{9} n^\eta$, the proof is complete. ■

Another problem of independent interest is counting the number of subsets of $\{1, 2, \dots, n\}$ of the form A^* where A is a multiset in $\{1, 2, \dots, n\}$. Szemerédi and Vu [14] recently solved this problem and showed that the number of sets A^* is $2^{O(\sqrt{n})}$.

3. Cycles in Hamiltonian Graphs

In order to prove [Theorem 1.2](#), we require a number of preliminary lemmas, which concern cycle structure in hamiltonian graphs. In the proof of [Theorem 1.2](#), it will be seen that the assumption of hamiltonicity may be made. In order to apply the results of [section 2](#), we will show that all but a negligible number of hamiltonian graphs G_n contain subgraphs which guarantee that $C(G_n)$ contains an η -large positive difference set or a translate of an η -large set of subset sums.

Throughout this section, we will assume that a hamiltonian graph G with a given hamiltonian cycle C is drawn in the plane in such a way that all the edges of G are straight line segments and the edges of C form a convex polygon.

Let G be a graph with a hamiltonian cycle C . Suppose G is a plane graph, C has k chords, and the bounded regions of G are $F_1, F_2, \dots, F_{k+1}$. Then G is called a k -ladder, denoted $H(k)$, if the boundaries of F_i and F_{i+1} share precisely one chord of C for $i = 1, 2, \dots, k$. The following lemma shows that such graphs guarantee a large positive difference set of cycle lengths:

Lemma 3.1. *If G contains a k -ladder, then $C(G)$ contains a translate of a positive difference set $(A - A)^+$ with $|A| = k + 1$.*

Proof. Suppose $H(k) \subset G$ and let $F_1, F_2, \dots, F_{k+1}$ denote the bounded regions of $H(k)$ such that the boundaries of F_i and the F_{i+1} share precisely one edge for $i = 1, 2, \dots, k$. Let a_i denote the number of edges on the boundary of F_i and let

$$A = \left\{ \sum_{i=1}^j (a_i - 2) : 1 \leq j \leq k + 1 \right\} \cup \{0\}.$$

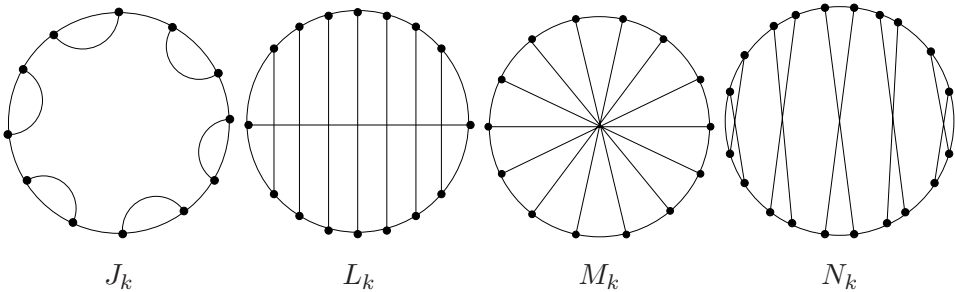
Then $C(G) \supset C(H(k)) \supset 2 + (A - A)^+$. ■

If G_n is a hamiltonian graph containing a vertex of degree $d \geq \sqrt{18n^{(1+\eta)/2}} \geq 2$, then $G_n \supset H(d-2)$. Therefore, by [Lemma 3.1](#), $C(G_n)$ contains an η -large positive difference set. This fact will be used later.

We now look at graphs which guarantee a translate of an η -large set of subset sums of cycle lengths. Let k be a positive integer and let C_m denote a cycle of length m labelled $v_0v_1 \dots v_{m-1}v_0$. Define the following graphs:

$$\begin{aligned} J_k &= C_{3k} \cup \{v_{3i}v_{3i+2} : 0 \leq i \leq k-1\} \\ L_k &= C_{8k+2} \cup \{v_0v_{4k+1}\} \cup \{v_iv_{8k+2-i} : 1 \leq i \leq 4k\} \\ M_k &= C_{8k+2} \cup \{v_iv_{i+4k+1} : 0 \leq i \leq 4k\} \\ N_k &= C_{4k} \cup \{v_{2i}v_{4k-2i-2}, v_{2i+1}v_{4k-2i-1} : 0 \leq i \leq k-1\}. \end{aligned}$$

Throughout this section, we will suppose that the vertices of J_k, L_k, M_k and N_k are labelled as in their definitions above. Label the hamiltonian cycle C_{3k}, C_{8k+2} or C_{4k} , respectively, C' . Natural drawings of J_k, L_k, M_k and N_k , with C' as a convex polygon, are illustrated below:



A graph G obtained by subdividing edges of C' in J_k, L_k, M_k and N_k is denoted $J(k), L(k), M(k)$ or $N(k)$, respectively.

Let G be a hamiltonian graph with hamiltonian cycle C . Chords e and f of C are *crossing* if C , together with e and f , is not a plane graph. A pair of crossing chords e and f of C are said to *cross trivially* if C , together with e and f , contains a 4-cycle through e and f . Otherwise e and f are said to *cross non-trivially*. If $N(k)$ contains no trivially crossing chords, then we relabel it $N^*(k)$. A *type k graph* is any of the graphs $J(k), L(k), M(k)$ or $N^*(k)$. The proof of the following lemma shows a reason for the additional requirement on $N(k)$:

Lemma 3.2. *Let G be a type k graph. Then $C(G) \supset a + A^*$, where A is a multiset in $\{1, 2, \dots, n\}$ of size at least k and a is a positive integer.*

Proof. Suppose $G = J(k)$. Let P_i be a $v_{3i}v_{3i+2}$ subpath of C for $i = 1, 2, \dots, k$ and suppose P_i and P_j are vertex-disjoint whenever $i \neq j$. Let $a_i = |P_i| - 1$

and set $a = |E(G)| - \sum_{i=1}^k (a_i - 1)$. Then, for any $S \subset \{1, 2, \dots, k\}$, we may find a cycle of length $a + \sum_{i \in S} (a_i - 1)$ in G . So $C(G) \supset a + A^*$ where A is the multiset with elements $a_1 - 1, a_2 - 1, \dots, a_k - 1$.

Suppose $G = L(k)$. Let P_i and Q_i be a pair of vertex-disjoint $v_i - v_{i+1}$ and $v_{8k+1-i} - v_{8k-i}$ paths, for $i = 1, 2, \dots, 4k - 1$. Let $x_i = |P_i| - 1$ and $y_i = |Q_i| - 1$ and $I = \{1, 3, \dots, 4k - 1\}$. Then, for any $S \subset I$, we may find a cycle in G through $v_0 v_{4k+1}$ which excludes the edges in $\bigcup_{i \in S} P_i$ and includes the edges in $\bigcup_{i \in S} Q_i$. Let a and b be the number of vertices in the two edge-disjoint $v_0 - v_{4k+1}$ subpaths of C . Then, for any $S \subset I$,

$$C(G) \supset a + \sum_{i \in S} (y_i + 2 - x_i).$$

Similarly, $C(G) \supset b + \sum_{i \in S} (x_i + 2 - y_i)$. For each $i \in I$, it is not possible that neither summand is positive. Without loss of generality, there exists $I' \subset I : |I'| \geq |I|/2 = k$ such that $a_i = x_i + 2 - y_i > 0$ for $i \in I'$. With $A = \{a_i : i \in I'\}$, we have $a + A^* \subset C(G)$ and $|A| \geq k$.

Suppose $G = M(k)$. Let P_i be a $v_i - v_{i+1}$ subpath of C , chosen so that P_i and P_{i+1} are edge-disjoint for $i = 0, \dots, 8k$. Let $x_i = |P_i| - 1$ and $y_i = |P_{i+4k+1}| - 1$ for $i = 0, 1, \dots, 4k$. Let $I = \{1, 3, \dots, 4k - 1\}$ and $S \subset I$. Then

$$C(G) \supset |P_0| + |P_{4k}| - 1 + \sum_{i \in S} (y_i + 2 - x_i)$$

and similarly $C(G) \supset |P_{4k}| + |P_{8k+1}| - 1 + \sum_{i \in S} (x_i + 2 - y_i)$. For $i \in I$, it is not possible that neither summand is positive. Without loss of generality, there is a set $I' \subset I : |I'| \geq |I|/2 = k$ such that $a_i = x_i + 2 - y_i > 0$ for $i \in I'$. With $a = |P_0| + |P_{4k}| - 1$ and $A = \{a_i : i \in I'\}$, $a + A^* \subset C(G)$ and $|A| \geq k$.

If $G = N^*(k)$, let P_i and Q_i denote the pair of vertex-disjoint $v_{2i-2} - v_{2i-1}$ and $v_{4k-2i} - v_{4k-2i+1}$ subpaths of C , respectively, for $i = 1, \dots, k$. Let $x_i = |P_i| - 1$ and $y_i = |Q_i| - 1$ and $a = |C| - \sum_{i=1}^k (x_i + y_i)$. Then, for any $S \subset \{1, 2, \dots, k\} = I$,

$$C(G) \supset a + \sum_{i \in S} (x_i + y_i - 2).$$

As no chords cross trivially, $a_i = x_i + y_i - 2 > 0$ for $i \in I$. So $C(G) \supset a + A^*$ where A has elements $a_1, a_2, \dots, a_k$. ■

It follows that if G_n contains a type k -graph and $k \geq 20n^\eta (\log_2 n)^2$, then $C(G_n)$ contains a translate of an η -large set of subset sums. The following lemma, which will be used to find such structure in hamiltonian graphs, generalizes a result due to Aldred and Thomassen [1], who proved it for $d = 1$:

Lemma 3.3. *Let k be a positive even integer and let G be a graph of maximum degree at most $d+2$, containing a hamiltonian cycle C . Suppose C has at least $(4k)^4d$ chords and each chord of C is non-trivially crossed by some other chord of C . Then G contains a type k -graph.*

Proof. Since any graph with m edges and maximum degree at most d contains a set of at least $m/2d$ independent edges, there is a set I of $128k^4$ independent chords of C . Partition C into k subpaths, each containing $256k^3$ end vertices of the edges in I . If for every subpath there is a chord of G with both ends in the subpath, then $J(k) \subset G$. Otherwise, choose a subpath P , containing no edge in I with both ends in P , and let the edges of I meeting P be $e_1, e_2, \dots, e_{256k^3}$, numbered according to the order in which their ends in P appear in some orientation of C . In the same orientation of C , these edges are re-ordered $e_{\sigma(1)}, e_{\sigma(2)}, \dots, e_{\sigma(256k^3)}$ according to the order of appearance of their ends on $C - V(P)$. If there exists an increasing sequence of $\sigma(i)$ of length at least $8k+2$, then $M(k) \subset G$. Otherwise, by an old result of Erdős and Szekeres, there exists a decreasing subsequence of numbers $\sigma(i)$ of length at least $256k^3/(8k+2) \geq (2k-1)(16k+4)$. Let $f(i) = u_i v_i$ denote the edges $e_{\sigma(i)} : 1 \leq i \leq (2k-1)(16k+4)$.

For $i=0, 1, \dots, 2k-2$, let

$$E_i = \{f((16k+4)i+j) : 1 \leq j \leq (16k+4)\} \text{ and} \\ V_i = \{u_j, v_j : f_j \in E_i\}.$$

Since every chord of C is non-trivially crossed by some other chord of C , for each $i: 0 \leq i \leq 2k-2$, $f((8k+2)(2i+1)) = e(i)$ is non-trivially crossed by some chord $e'(i)$ of C . If some end of $e'(i)$ is not in V_i , then $L(k) \subset G$. So we suppose the ends of $e'(i)$ are in V_i for $i=0, 1, 2, \dots, 2k-2$. If both ends of $e'(i)$ are not in P for at least k values of i , then $J(k) \subset G$. Therefore we suppose $e'(i)$ has one end in P and one end in $V_i \setminus V(P)$ for $i \in I \subset \{0, 1, 2, \dots, 2k-2\}$ with $|I| \geq k$. The edges $e(i)$ and $e'(i)$ for $i \in I$, together with C , now comprise $N^*(k)$ in G . ■

The next lemma deals with hamiltonian graphs do not necessarily satisfy the hypotheses of [Lemma 3.3](#):

Lemma 3.4. *Let k be a positive integer and let G be a graph of maximum degree at most $d+2$. Suppose G contains a hamiltonian cycle C with m chords. Then either*

- (1) G contains a type k graph or
- (2) G contains a plane graph containing C and at least $\frac{1}{2}(m - (4k)^4)$ chords of C .

Proof. Colour a chord of C blue if some chord of C non-trivially crosses it, and red otherwise. Then the set of blue edges and the edges of C span a subgraph of G satisfying the hypotheses of [Lemma 3.3](#). We therefore assume that G contains at most $(4k)^4 d$ blue edges and at least $m - (4k)^4 d$ red edges. The set of red edges, together with the edges of C , span a graph in which any crossing edges cross trivially. So we may remove a set of at most half the red chords of C to obtain a plane subgraph of G satisfying (2). ■

The following elementary lemma will be used in the next corollary:

Lemma 3.5. *Let T be a tree. Then T contains at least s leaves or a path of length at least $|T|/2s - 2$.*

Proof. If T contains at most s leaves, then T contains at least $|T| - 2s$ vertices of degree two. This implies that some set of at least $|T|/2s - 1$ vertices of degree two in T induce a path in T . ■

Corollary 3.6. *Let G satisfy the hypotheses [Lemma 3.3](#). Then G contains a type k graph or $G \supset H(r)$ where $r \geq \max\{\frac{1}{2k}(m - (4k)^4 d) - 2, d\}$.*

Proof. If G does not contain a type k graph then, by (2) in [Lemma 3.4](#), G contains a plane graph F containing C and at least $\frac{1}{2}(m - (4k)^4 d)$ chords of C . By Euler's formula, F has at least $\frac{1}{2}(m - (4k)^4 d)$ bounded faces. The vertices of the plane dual graph of F corresponding to these bounded faces induce a tree T . By [Lemma 3.5](#), T contains at least k leaves or a path of length at least $\frac{1}{2k}(m - (4k)^4 d) - 2$. If the former case, $J(k) \subset F$ and in the latter case, C , together with the edges on the boundaries of faces in F corresponding to vertices of P , forms an $H(r)$, where $r \geq \frac{1}{2k}(m - (4k)^4 d) - 2$. By [Lemma 3.1](#), G also contains an $H(d)$. This completes the proof. ■

This corollary is fundamental in proving [Theorem 1.2](#).

4. Proof of [Theorem 1.2](#)

Using the results of [section 2 and 3](#), we prove [Theorem 1.2](#):

Proof of [Theorem 1.2](#). If $\mathcal{H}$ is any class of graphs, we write $C_{\mathcal{H}}(n)$ for the number of cycle sets on $\{1, 2, \dots, n\}$ arising from graphs on n vertices in $\mathcal{H}$. When $\mathcal{H}$ is the class of all graphs, we write $C(n)$ instead of $C_{\mathcal{H}}(n)$. If $\mathcal{H}_1$ is the class of graphs of circumference less than $n - n^{1/10}$, then clearly $\log_2 C_{\mathcal{H}_1}(n) < n - n^{1/10}$. Let $\mathcal{H}_2$ be the class of graphs of circumference at least $n - n^{1/10}$ containing less than $n + n/(4 \log_2 n)$ edges. Let G_n be a graph

of order n in $\mathcal{H}_2$. Fixing a cycle of length at least $n - n^{1/10}$ in G_n , the number of ways of placing the remaining edges is at most

$$\binom{n}{2}^{n/(4\log_2 n) + n^{1/10}} = 2^{n/2 + o(n)}.$$

This is an upper bound for $C_{\mathcal{H}_2}(n)$. Now let $\mathcal{H}_3$ denote the class of graphs not in $\mathcal{H}_1$ or $\mathcal{H}_2$. We will show that if $G_n \in \mathcal{H}_3$, then G_n contains a $\frac{1}{10}$ -large positive difference set or a translate of a $\frac{1}{10}$ -large set of subset sums.

Let C be a longest cycle in a graph $G_n \in \mathcal{H}_3$. If some vertex $v \in V(G_n) \setminus V(C)$ has at least $\sqrt{18}n^{2/3} + 2$ neighbours on C , then we easily find a cycle C' in G_n containing v and all the neighbours of v . By [Lemma 3.1](#), this implies $C(G_n)$ contains a $\frac{1}{3}$ -large positive difference set. If no vertex of $G_n - V(C)$ has at least $\sqrt{18}n^{2/3} + 2$ neighbours on C , then C has at least

$$n + n/(4\log_2 n) - n^{1/10}(\sqrt{18}n^{2/3} + 1) - n^{1/5} = n + \Omega(n/\log_2 n)$$

chords. By [Corollary 3.6](#), with $k = 20n^{2/19}(\log_2 n)^2$ and $d = \sqrt{18}n^{1/2+1/19} + 2$, H contains a type k -graph or $H(r)$, where $r = \max\{d - 2, \frac{1}{2k}(\Omega(n/\log_2 n) - (4k)^4 d)\}$. By the choice of d and k , this maximum is $d - 2 = n^{1/2+1/19}$, so G contains $H(d - 2)$. By [Lemma 3.1](#) and [Lemma 3.2](#), this implies that $C(H)$ contains a $\frac{2}{19}$ -large positive difference set or a translate of a $\frac{2}{19}$ -large set of subset sums. By [Lemma 2.3](#) and [Lemma 2.5](#), the number of subsets of $\{1, 2, \dots, n\}$ containing a $\frac{2}{19}$ -large positive difference set or a translate of a $\frac{2}{19}$ -large set of subset sums is $O(2^{n-n^{2/19}})$. The proof of [Theorem 1.2](#) is now complete with $c \geq \frac{1}{10}$. ■

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